



Figure 3: appEZ components

Unified integration components (UIC)

UIC enables users to create multi-platform native mobile applications with simple HTML5/CSS3 and JavaScript. This facilitates the creation of cross-platform applications with a unified HTML structure. These components are built on open source industry standards, frameworks and tools such as LESS (for CSS programmability), Bootstrap (for page layouts and dynamic structuring), and JQuery Mobile (for gesture handling and navigation history management).

SmartWeb

This is a JavaScript based model-view-controller (MVC) skeleton for the business logic layer. SmartWeb recommends a generic design skeleton that can be used to develop typical client-server applications. Its MVC skeleton for Web development supports scalability and extensibility as per application business logic layer requirements.

Mobilet Manager Interface (MMI)

This is an interface to communicate between the native layer and the Web layer. MMI provides a set of APIs that enables developers to leverage native capabilities exposed by native mobile platforms (Android/iOS/WP) from the JavaScript layer. MMI has the following parts:

- *MMI (JavaScript)*: Provides a defined set of APIs for the developers to use at the Web layer in JavaScript.

- *MMI (native)*: This is the corresponding native layer at the respective platform container that receives the required parameters from the Web layer and executes native services as per the need.
- *MMI callback notification*: Each MMI API requires callback at the Web layer to get notified about success/failure from native container layer (Android/iOS/WP). appEZ supports the Android (API level 10 onwards), iOS6 - iOS9 and Windows Phone 8/8.1 mobile platforms for mobile application development. It has some advantages over competing frameworks. These include:
 - Unique offering of flexible architecture for true mobile hybrid application development
 - Flawless amalgamation of native and hybrid technologies as per application's needs
 - Covers all the layers of mobile hybrid application development
 - Modular architecture makes it possible to use only the required components as per business requirements, as against using the complete library set-up
 - Ready-to-use services for accessing the most commonly used features of the native container
 - Expressive and easy-to-use JS syntaxes
 - Based on open source libraries which makes the design/development extensible 

References

- [1] 'Enterprise Mobility Breakthrough', < <http://www.amazon.com/Enterprise-Mobility-Breakthrough-Beginners-Guide/dp/1482844087>, https://books.google.com/books/about/Enterprise_Mobility_Breakthrough.html?id=sY5rqEACAAJ&hl=en >
- [2] 'appEZ', < <http://appez.github.io/> >

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